ASSIGNMENT NO. 2

COMPARATIVE STUDY AND IMPLEMENTATION OF GENERATIVE AI MODELS

Image Reconstruction and Image Generation

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INTRODUCTION & MODELS STUDIED

Generative AI learns patterns from data to rebuild or create images. Here we compare two ways of working: reconstructing an input image, and generating a new one.

IMAGE RECONSTRUCTION

Autoencoder (AE)

An encoder squeezes the image into a small latent code; a decoder rebuilds the image. Trained to minimise MSE.

Image → Encoder → z → Decoder → Image

Variational Autoencoder (VAE)

The encoder predicts a mean and variance; a sampled latent z is decoded. Loss = reconstruction + KL divergence.

Image → μ, σ → sample z → Decoder

IMAGE GENERATION

DCGAN

A generator turns random noise into images while a discriminator learns to tell real from fake.

Noise → Generator → Fake image ⇄ Discriminator

Stable Diffusion (pretrained)

A pretrained latent diffusion model removes noise step by step, guided by a text prompt.

Prompt → Text encoder → Denoising → Image

AE, VAE and DCGAN use MNIST digits; Stable Diffusion generates images from text prompts.

IMPLEMENTATION & EXPERIMENTAL SETUP

Environment

- Google Colab notebook, NVIDIA Tesla T4 GPU (CUDA)
- Python 3.13.16, PyTorch 2.11.0+cu130, torchvision 0.26.0+cu130
- Random seed 42 for repeatable runs

Data and models

- MNIST subset: 12,000 train / 2,000 test images
- AE, VAE, DCGAN: 10 epochs each, batch size 128
- Latent size: AE 32, VAE 16; DCGAN noise 64
- Stable Diffusion v1.5: pretrained, not trained here; 20 steps, guidance 7.0, 512 × 512

DIFFERENT INPUTS AND OUTPUTS

AE

MNIST image → AE → reconstructed digit

VAE

MNIST image → VAE → reconstruction; random z → new digits

DCGAN

Random noise → generator → digit-like image

Stable Diffusion

Text prompt → pretrained pipeline → 512 × 512 image

RESULTS & COMPARATIVE ANALYSIS

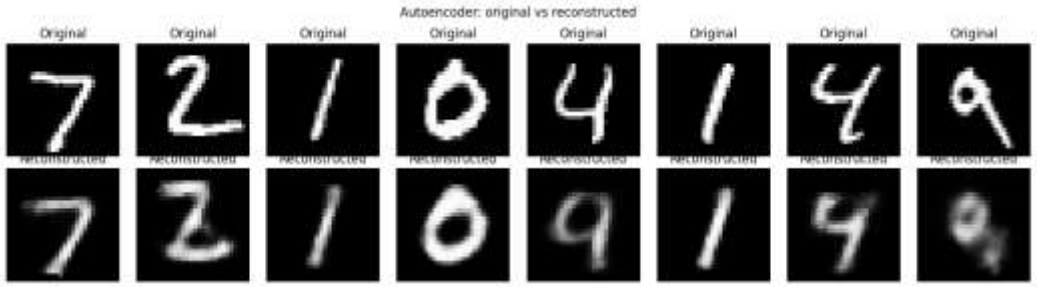


Fig. 1 AE: original (top) vs reconstructed (bottom)

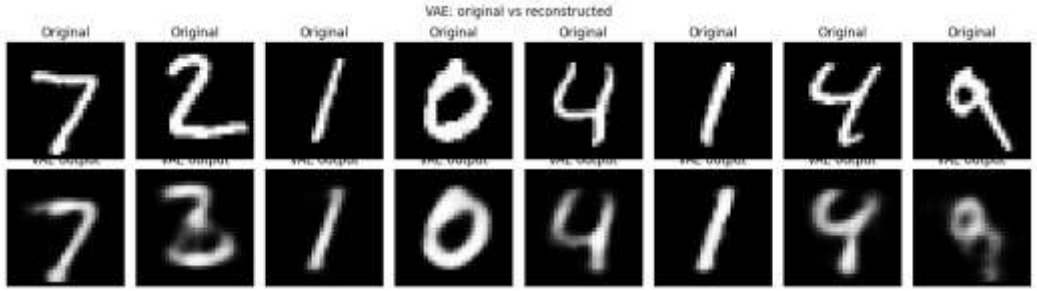


Fig. 2 VAE: original (top) vs reconstructed (bottom)



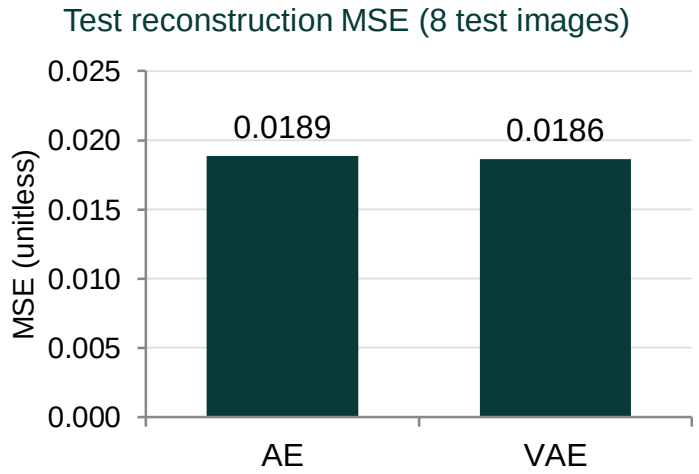
Fig. 3 DCGAN samples



Fig. 4 Stable Diffusion

Prompts: tiger, kingfisher, mountain sunrise, wildflowers

Model	Input → output	Observed in notebook outputs
AE	MNIST digit → reconstruction	Digit shapes kept; strokes blurred; one 4 drawn closer to a 9
VAE	Digit → reconstruction; random z → samples	Similar blur; the 2 distorted; random samples often malformed
DCGAN	Noise → digit-like image	Digit-like strokes; many artefacts; some not recognisable
Stable Diffusion	Text prompt → 512 × 512 image	Detailed, realistic scenes; minor artefacts (e.g. stray branch piece)



AE train loss (MSE): 0.0895 → 0.0222

VAE train loss (BCE+KL): 248.56 → 119.69

Losses use different definitions, so they are not compared. Test MSE uses only 8 test images: not a ranking.

CONCLUSION & LEARNING OUTCOMES

Model	Strength	Limitation
AE	Simple; compact latent code	Blurs details; latent space not structured for sampling
VAE	Probabilistic latent space; can sample	Blurry outputs; random samples often unclear
DCGAN	Sharp-edged digit-like samples	Artefacts after short training; training can be unstable
Stable Diffusion	Detailed, text-guided images with no training here	Heavy compute; prompt details not always followed

Key learning

- Reconstruction rebuilds a given input; generation creates new images
- A pretrained diffusion model gives text-conditioned detail without training
- No single model wins: each suits a different goal

Study limitations

Different data and outputs (MNIST digits vs text-to-image), small subsets, 10 epochs, one run: a capability comparison, not a benchmark

Applications

AE/VAE: compression, denoising, anomaly detection. DCGAN: data augmentation. Stable Diffusion: concept art, design drafts.